

Total
/35

Name _____

Date _____

KNOWLEDGE

1. Find the general solution to each of the following differential equations.

a. $\frac{dy}{dx} = 6x^2 + 20 \cos(5x) - \sec^2 x$

b. $\frac{dy}{dx} = \frac{3}{x} - 4e^{2x} \sin(e^{2x})$

(1)

(1)

2. Solve the following initial value problems explicitly.

a. $\frac{dy}{dx} = 9x^2 + 6e^{3x+9}, \quad y = 7 \text{ when } x = -3$

b. $\frac{dy}{dx} = y^2 \sin x, \quad y = 2 \text{ when } x = \pi$

(2)

(2)

3. Use substitution to evaluate the integral.

a. $\int e^x \sqrt{1 + e^x} dx$

b. $\int \frac{dt}{\cos^2 t \sqrt{1 + \tan t}}$

(2)

(2)

4. Construct a slope field for the differential equation, $\frac{dy}{dx} = y - 2x$. Then use it to sketch a solution curve that passes through the given point (1,0),

(3)

APPLICATION

5. How long will it take an investment to double in value if the interest rate is 6% compounded continuously?

(2)

6. A roast turkey is taken from an oven when its temperature has reached 185°F and is placed on a table where the temperature is 75°F.

a. If the temperature of the turkey is 150°F after half an hour, what is the temperature after 45 minutes?

(3)

b. When will the turkey have cooled to 100°F?

(2)

7. The half-life of cesium-137 is 30 years. Suppose we have a 100-mg sample.

a. Write an equation to represent the decay of Cesium-137.

(2)

b. How much of the sample remains after 100 years?

(1)

c. After how long will only 1 mg remain?

(2)

COMMUNICATION

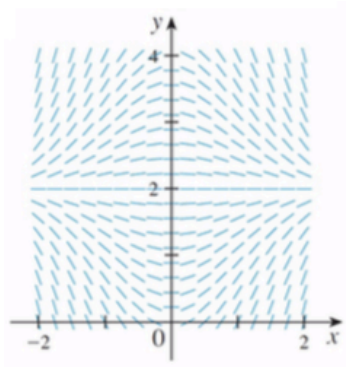
8. a. For each of the slope fields shown below, write the letter of the corresponding differential equation.

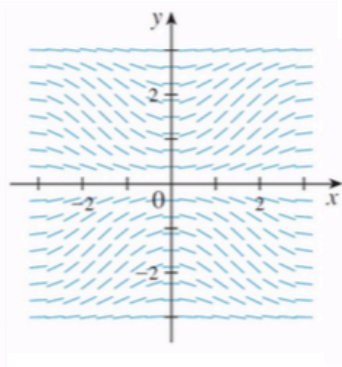
A) $\frac{dy}{dx} = 2 - y$

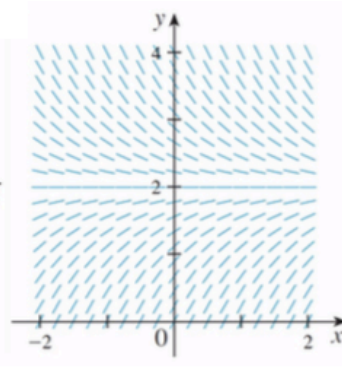
B) $\frac{dy}{dx} = x(2 - y)$

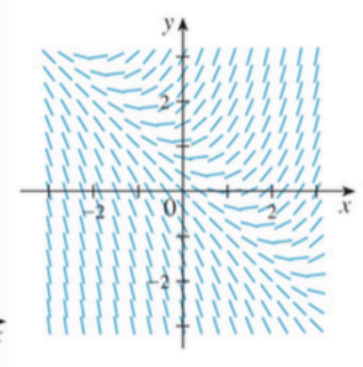
C) $\frac{dy}{dx} = x + y - 1$

D) $\frac{dy}{dx} = \sin x \sin y$









(2)

b. Justify/explain each of your responses from part a.

(2)

THINKING

9. Evaluate $\int \frac{8-w}{4w^2+9} dw$

(3)

10. Evaluate $\int \sin^3 x \sqrt{\cos x} dx$.

(3)